

NEEDS ANALYSIS

Tailboard Talk 2.0: Job Briefings for a Modernizing Grid

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Project Overview

This is a proof-of-concept microlearning module developed to demonstrate instructional design process against a real, current strategic priority. It is a portfolio piece, not an actual Hydro Ottawa work product: the strategic and regulatory context below is drawn from Hydro Ottawa’s own public materials and general electrical-utility safety regulation, but the specific field procedures depicted (lockout confirmation steps, distributed energy resource disconnect verification) are illustrative and modeled on general industry practice, not Hydro Ottawa’s actual internal procedures, which I don’t have access to as a candidate.

Business / Strategic Driver

Hydro Ottawa’s 2026-2030 Strategic Direction names grid modernization as its lead priority: nearly doubling grid investment to build smart-grid capability, including remote-controlled switching, automated fault detection, and automated system restoration, alongside growing integration of distributed energy resources (rooftop solar, battery storage) and “prosumer” customers who both consume and generate power. Net-zero and sustainability commitments (via Portage Power and Envari) point the same direction: more distributed generation on the grid, sooner rather than later.

This is a well-documented, industry-wide pattern not unique to Hydro Ottawa: as utilities modernize, field crews face a genuinely new hazard category. A line can now potentially be re-energized by an automated device rather than a person, or backfed by a customer’s own generation, in addition to the traditional single source (the system operator). Existing safety-communication habits, built around a single source of energization, need an explicit update to keep pace with the infrastructure itself.

Performance Problem / Gap Statement

The gap is not that field crews don’t know how to run a tailboard talk because, under OHSA and IHSA safe-practice guidelines, that process (hazards, protective measures, roles, confirmed via a request-confirm-release clearance chain with the system operator) is already thoroughly trained and regulated. The gap is that the standard tailboard-talk checklist hasn’t explicitly caught up to modernization: without a deliberate update, an experienced crew’s well-practiced habits can carry a “one energization source, one clearance tag” assumption that no longer holds on a modernized circuit.

Target Audience

Audience	Prior Knowledge	Gap This Module Addresses
Primary: field crew / powerline technicians	Fully trained on OHSA/IHSA switching-and-clearing and tailboard-talk procedures; multiple years of field experience	Has not yet had the tailboard-talk checklist explicitly updated for automated switching and DER backfeed as modernization rolls out

Secondary: field supervisors who lead briefings	Same baseline as above, plus responsibility for running the briefing and confirming crew understanding	Needs to model and enforce the updated question set, not just personally know it
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Task / Content Analysis

Standard tailboard talk today	Updated: Tailboard Talk 2.0
1. Confirm clearance tag / isolation with system operator	1. Confirm clearance tag / isolation with system operator (unchanged)
2. Review hazards specific to the job	2. Review hazards specific to the job (unchanged)
3. Confirm protective measures (grounding, PPE)	3. Confirm protective measures (grounding, PPE) (unchanged)
4. Confirm each person's role	4. Confirm each person's role (unchanged)
— (not currently a standard step)	5. NEW: Confirm: can this circuit be re-energized by an automated/remote switching device, or backfed by a distributed energy resource? If yes, confirm manual lockout and DER disconnect verification specifically, not just the standard tag.

Learning Objectives

Written to the ABCD (Audience–Behavior–Condition–Degree) format used throughout Hydro Ottawa's posting language around measurable, results-driven outcomes:

1. Given a job-site scenario involving a circuit with automated switching devices or nearby distributed energy resources, the learner will correctly identify that an additional clearance-confirmation question is required before work begins, with no incorrect identifications on the graded scenario.
2. The learner will state, in their own words, the two new hazard categories introduced by grid modernization (automated/remote re-energization; DER backfeed) with 100% accuracy on the knowledge check.
3. The learner will apply the updated tailboard-talk job aid to a simulated pre-job briefing, correctly sequencing hazards, protective measures, roles, and the new modernization check.
4. (Transfer objective, evaluated on the job, not in the module) The learner will use the Tailboard Talk 2.0 job aid on their next work order involving a modernization-affected circuit.

Delivery Recommendation

On-demand, mobile-enabled microlearning built in Articulate Rise 360 matching the tools and delivery modes named directly in the posting. Estimated learner runtime: 10–12 minutes, short enough to assign ahead of a smart-grid device rollout in a given service area without pulling crews off the schedule. The experiential core is a branching scenario simulating a real tailboard-talk conversation, so learners practice the judgment call, not just recall a fact. A companion one-page PDF job aid is designed for printing and keeping in the truck, since field connectivity for a mobile device can't be assumed.

Constraints

- Built as a proof-of-concept scoped to one short module rather than a full training program.
- No access to Hydro Ottawa's actual internal switching procedures, SME review, or brand assets; content is deliberately built on public strategic materials and general industry/regulatory practice, and is framed as illustrative throughout.

- Single-person production (no dedicated videographer or voice talent).

Evaluation Plan (Kirkpatrick-Aligned)

Level	In this proof-of-concept	In a real deployment
1. Reaction	N/A (single-audience demo)	Post-module micro-survey: relevance and clarity ratings
2. Learning	Knowledge check + scenario completion, built into the module	Same, tracked via LMS/SCORM at scale
3. Behavior	Job aid designed for field use; not field-tested here	Spot audits of tailboard talks on modernization-affected circuits
4. Results	N/A	Trend in near-miss/incident reports tied to automated switching or DER backfeed

Sources Consulted

- Hydro Ottawa, “Past progress, future powered: Hydro Ottawa highlights record 2025 investments and launches 2026-2030 Strategic Direction” (newswire.ca)
- Hydro Ottawa, “Grid modernization” — 2026-2030 investment plan (hydroottawa.com)
- Hydro Ottawa, “Hydro Ottawa commits to being net-zero by 2030” (hydroottawa.com)
- General framework: Occupational Health and Safety Act (Ontario); Infrastructure Health and Safety Association (IHSA) safe-practice guidelines for line clearing and switching/clearing procedures